



Morphology-Aware Identification of Shared Bicycle Parking Hotspots for Urban Public-Space Management: Evidence From Guangzhou

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Abstract: The rapid expansion of dockless shared bicycle systems creates persistent challenges for parking management and the efficient use of urban public space, particularly around transport hubs and major commuting corridors. Conventional hotspot-identification methods mainly indicate where parking demand is concentrated but provide limited information on the spatial form and potential public-space impact of bicycle accumulation. This study develops a morphology-aware framework using shared bicycle order and trajectory data from Guangzhou. After removing operational repositioning events, dense parking cores were extracted through grid-based connected components. Principal component analysis and robust spread measures were then used to quantify cluster orientation, length–width ratio, and bandwidth. Road proximity and orientation consistency were incorporated to distinguish roadside-band (RB), intersection-corner (IC), designated-parking-area-like (DPA), and scattered points (SP) morphologies. Short-duration demand surges during morning and evening peaks were identified separately and combined with high demand and band-shaped morphology to screen priority governance locations. Analysis of 6.902 million orders identified 27,687 dense parking cores, with the top 1% accounting for 43.7% of within-cluster orders. Band-shaped clusters represented 13.4% of all cores but accounted for 32.1% of orders. Among 26,142 strictly classified cores, RB clusters accounted for 3.23%, and only 21.8% were aligned within 30° of adjacent roads, indicating that oblique and multi-row occupation was more common than simple linear roadside parking. Among 1,842 high-demand cores, 320 showed morning-peak surges and 80 evening-peak surges. High-demand locations within 200 m of metro entrances recorded 1.7 times the mean order volume of those outside the metro buffer. The combined surge–demand–morphology criterion identified 240 priority governance points, 64.6% near arterial or sub-arterial roads and 50.0% near designated parking facilities. The findings show that parking pressure is shaped by both demand intensity and accumulation morphology, providing a practical basis for differentiated parking management, pre-peak dispatch, and improved use of urban public space.

Keywords: Shared bicycles; Parking management; Urban public space; Parking hotspots; Morphology-aware clustering; Active mobility; Urban mobility governance; Spatial decision support

1 Introduction

Shared bicycles have become an important component of urban last-mile mobility, particularly in large cities where they connect residential areas, employment centres, and public transport nodes. Their dockless operating model, however, also creates a persistent management problem: bicycle accumulation is highly mobile, spatially uneven, and often concentrated in public spaces that must simultaneously accommodate pedestrians, cyclists, public transport users, and other street activities. The problem becomes especially visible during short periods of intense accumulation. During the morning peak, bicycles tend to converge around rail transit stations and employment districts, whereas during the evening peak they move back toward residential areas. At many locations, this produces elongated or multi-row parking along roadsides and pedestrian spaces. Because these accumulations can emerge

within a short period, post-hoc clearance alone is often unable to respond before local parking pressure becomes severe. For urban management, the more useful question is therefore not simply where bicycle use is high across the city, but when concentrated accumulation occurs, where it occurs, and what spatial form it takes. Identifying these three dimensions together is necessary for more timely dispatch, parking management, and allocation of street and public space.

Guangzhou provides a particularly relevant setting for examining this problem. The number of registered electric bicycles in the city exceeded 6.51 million in 2025, accounting for 17.9% of all-mode travel and creating substantial competition for non-motorised road space with shared bicycles. Guangzhou has consequently regulated the total number of shared bicycles at 300,000–480,000 for 2025–2028, while the quotas for Qingju, Meituan, and Hello in the central six districts were approximately 99,000, 109,000, and 80,000, respectively, in the second quarter of 2026. Around metro stations in particular, competition for limited curbside and pedestrian space between shared bicycles and electric bicycles has become increasingly visible. Measures such as “one station, one policy” and the provision of additional parking areas have reduced some local conflicts, but several operational tasks remain dependent on more precise spatial evidence. Electronic-fence optimisation, no-parking-zone delineation, pre-peak dispatch, and the management of existing parking facilities all require information not only on parking intensity but also on the morphology of accumulation, including roadside bands, corner concentrations, and designated parking areas. This relationship between bicycle accumulation and the management of limited urban public space forms the practical starting point of the present study.

Research on shared bicycle systems and travel demand has developed along several complementary directions. At the system level, the evolution of public bicycle systems in Europe, the Americas, and Asia has been extensively reviewed [1, 2], together with issues related to vehicle supply, usage behaviour, and their influencing factors [3, 4]. Changes in residents’ travel behaviour following the introduction of public bicycle systems have also been examined in Shanghai [5], while broader planning challenges associated with bicycle development in urban China have received increasing attention [6]. At the facility and policy levels, a range of measures for promoting cycling have been synthesised [7]. Environmental impacts constitute another important research stream, with large-scale data being used to assess energy-saving and emission-reduction effects as well as life-cycle carbon emissions [8, 9]. A second major direction concerns the spatiotemporal distribution of bicycle demand. Spatial and temporal data have been widely used to reveal bicycle-sharing usage patterns [10, 11], while subsequent studies have examined parking-space identification, spatiotemporal usage patterns, tidal characteristics, and peak-period demand patterns [12–15]. The effects of the built environment, weather conditions, and calendar factors on bicycle demand have likewise been investigated extensively [16–21]. Periodic characteristics of system operation have been identified in several studies [22–24], and deep-learning approaches have increasingly been introduced for short-term demand forecasting at station and grid scales [25–27]. Collectively, this body of research provides a strong basis for identifying when and where bicycle demand occurs. However, demand is usually represented by order volume, rental frequency, or temporal intensity, which does not necessarily indicate how parking occupies urban space. A high-demand location may function efficiently within a designated parking area, whereas a smaller accumulation may cause greater disruption if bicycles extend across a narrow sidewalk or roadside space. Demand intensity and spatial impact therefore represent related but distinct dimensions of the same urban management problem.

The identification of parking accumulation has most commonly relied on density-based spatial clustering. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [28], Ordering Points To Identify the Clustering Structure (OPTICS) [29], and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) [30] are well suited to identifying non-convex groups of spatial observations, but their primary output is cluster membership rather than an explicit description of within-cluster morphology. Consequently, clusters with substantially different implications for street-space management may be treated in similar ways. An elongated roadside band, an intersection-corner accumulation, and a compact designated parking area may contain comparable numbers of observations while occupying public space in fundamentally different ways. The applicability and limitations of DBSCAN for large datasets have been systematically discussed [31], although morphological interpretation of the resulting clusters has generally remained outside the scope of such work. In shared micromobility research, spatial clustering and hotspot-analysis methods have been applied to identify parking aggregation areas, inconsiderate-parking patterns, and bicycle-sharing hotspots [32–34]. Usage patterns, parking characteristics, and their influencing factors in dockless bicycle systems have also been investigated [35, 36]. Geo-fence planning has further been explored as a means of organising dockless bicycle parking space and mitigating inappropriate parking [37]. More recent work has increasingly focused on disorderly parking and the occupation of public space by shared micromobility systems. In particular, the spatiotemporal heterogeneity of inconsiderate parking and its relationship with the built environment has been examined [38], while spatial clustering has also been used to identify potential geo-fenced parking spaces for free-floating bicycles [12]. Related studies have further applied hotspot identification and classification under electric-fence conditions [34]. Together, these studies demonstrate the value of spatial clustering and hotspot analysis for locating parking concentrations and supporting parking management. However, the internal morphology of

individual parking clusters remains insufficiently represented. In particular, distinctions among elongated roadside accumulation, intersection-corner parking, compact designated parking areas, and dispersed parking points have received comparatively limited attention as explicit post-clustering dimensions. This limits the extent to which hotspot identification can be translated into differentiated urban-management responses.

A related body of work provides useful analytical foundations for linking parking morphology with the urban street network. Spatial partitioning methods have been developed for urban transportation networks by incorporating congestion patterns while preserving the spatial compactness of resulting clusters [39]. Another research stream concerns the functional integration of bicycles with public transport. The determinants of bicycle–metro integration have been examined in Beijing [40], while bicycles have also been discussed as a feeder mode for public transport [41]. The broader integration of shared bicycles and public transit has likewise been evaluated [42]. These studies help clarify where bicycle demand is associated with public transport, but they do not directly address the geometry of parking point sets or the relationship between cluster orientation, road alignment, and curbside occupation. Bicycle-system operation and dispatch have developed along a separate line of research. System design under service-level constraints has been optimised [43], while static rebalancing and vehicle-routing approaches have been proposed to improve operational efficiency [44, 45]. Dynamic demand-driven clustering has also been introduced to support bicycle repositioning operations [46]. Although these approaches are important for fleet management, the spatial form of parking accumulation is generally treated as secondary to demand volume and routing efficiency. Linear-pattern extraction and road-orientation analysis have received comparatively limited attention in the automatic interpretation of shared bicycle parking morphology. Integrating road-network orientation with the geometric structure of parking clusters therefore provides a way to connect spatial identification with curbside occupation, public-space pressure, and location-specific management.

Three research gaps emerge from the above literature. First, existing hotspot-identification methods usually determine whether observations belong to a dense cluster but provide limited information on the internal geometry of that cluster. This makes it difficult to distinguish morphologically different forms of parking accumulation and, consequently, to determine whether a hotspot represents compact and relatively orderly parking or an elongated form that is more likely to occupy roadside or pedestrian space. Second, morphology, functional location, and demand intensity are often combined within a single hotspot concept. Treating these dimensions separately is important because a location can have high demand but limited spatial impact, or comparatively moderate demand but substantial public-space occupation. Without such separation, the connection between hotspot identification and differentiated management remains weak. Third, commonly used demand indicators are often based on total order volume over relatively long periods. They therefore provide limited information on short-duration accumulation during commuting peaks, although these brief periods are precisely when public-space conflicts and dispatch requirements are most acute.

To address these gaps, this study develops a morphology-aware framework for identifying shared bicycle parking hotspots and translating the results into point-level urban management information. The contribution is threefold. First, parking morphology is treated as an independently measurable property after cluster extraction. Principal component analysis (PCA) and a robust spread measure based on the 5th–95th percentile range are used to quantify cluster length–width ratio, bandwidth, and major-axis orientation. These geometric measures are combined with road-proximity distance and orientation consistency to distinguish four morphology types: roadside-band (RB), intersection-corner (IC), designated-parking-area-like (DPA), and scattered points (SP). Short-axis bandwidth is further used to approximate the number of parallel parking rows and to describe multi-row occupation. Second, parking demand is represented as a short-duration surge rather than solely as total order volume. Morning- and evening-peak order shares are used to identify temporally concentrated accumulation associated with contexts such as metro access and commuting. This allows demand intensity to be examined separately from parking morphology and estimated spatial impact. Third, the two dimensions are brought together through a “short-duration surge $\times$ high-demand $\times$ band accumulation” criterion to identify priority governance locations. The resulting point list is then interpreted using area of interest (AOI)/point of interest (POI) relationships to distinguish locational contexts and to support differentiated responses, including pre-peak dispatch, parking-area management, and the more targeted use of urban street and public space.

2 Methodology

2.1 Overall Analytical Framework

The analytical framework integrates shared bicycle order and trajectory data with urban road networks, functional zones, and transport and public-service facilities. It was designed to distinguish three aspects of bicycle parking that are directly relevant to urban management: the spatial form of accumulation, the timing and intensity of concentrated demand, and the resulting priority for intervention. In this study, a parking hotspot refers to a spatially aggregated dense parking core that satisfies the clustering criteria and is subsequently characterized by morphology and demand indicators.

The analysis proceeded in three stages. First, bicycle parking events were processed to identify dense spatial cores and quantify their geometric characteristics. These characteristics were then linked to the surrounding road network to construct a morphology-labelled parking hotspot database. Second, temporal information from bicycle orders was combined with AOI, POI, metro-entrance, and bus-stop data to identify short-duration demand surges and characterize the locational context of the corresponding clusters. Third, parking morphology and short-duration demand were jointly evaluated to identify locations where concentrated use was accompanied by potentially disruptive parking forms. These locations were then classified according to their surrounding facilities and road environment to support differentiated parking-management responses. The overall analytical framework is shown in Figure 1.

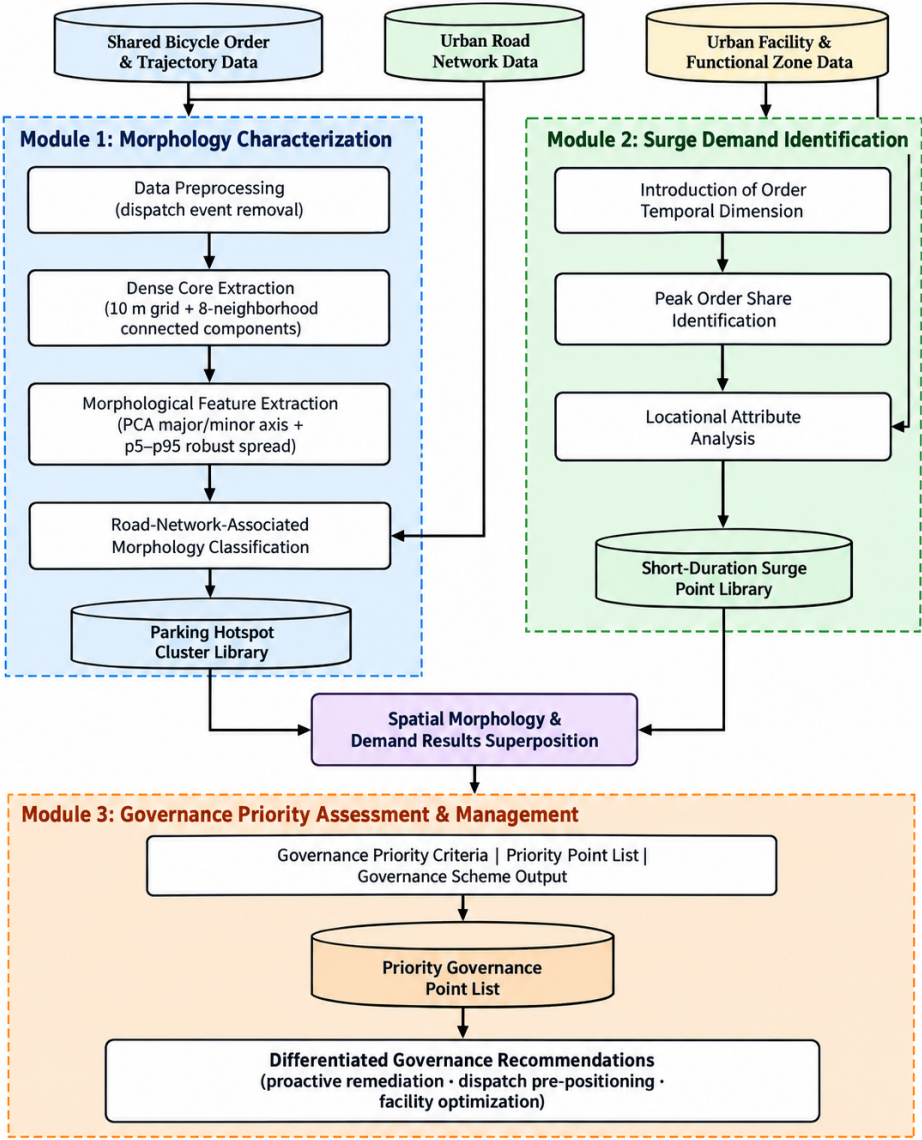


Figure 1. Analysis framework of morphology-aware clustering for shared bicycle parking hotspot identification
 Note: PCA = principal component analysis.

(1) Characterization of spatial morphological features of parking hotspots

Cleaned bicycle order and trajectory data were first used to extract dense parking cores. Morphological features were then calculated for each core, followed by classification in relation to the road network. PCA was used to determine the dominant direction of each cluster, while the 5th–95th percentile range of the projected coordinates was used as a robust spread measure to quantify its longitudinal extent, lateral bandwidth, and length–width ratio. These geometric measures were combined with road-network characteristics to distinguish four parking morphologies: RB, IC, DPA, and SP. This step transformed large numbers of discrete parking events into spatial units that could be interpreted in terms of their form and potential occupation of street and public space.

(2) Identification of short-duration demand-surge clusters

Temporal information was then introduced into the morphology-labelled hotspot database. Morning- and

evening-peak order shares were calculated to identify clusters where demand was concentrated within relatively short periods. AOI, POI, metro-entrance, and bus-stop data were subsequently used to describe the locational context of these clusters. Particular attention was given to accumulation associated with metro access, employment areas, commercial activity, and residential commuting. The resulting analysis distinguished locations with sustained high use from those experiencing short-duration peak-period surges.

(3) Governance priority assessment and differentiated management recommendations

In the final stage, spatial morphology and short-duration demand were jointly assessed. A “short-duration surge $\times$ high-demand $\times$ band accumulation” criterion was used to identify locations where concentrated bicycle demand coincided with parking forms more likely to interfere with street or public-space use. The resulting priority locations were further interpreted in relation to surrounding AOI/POI characteristics and road hierarchy. This provided a point-level basis for distinguishing different management situations and for formulating responses such as pre-peak dispatch, parking guidance, remediation of roadside accumulation, and improved use of existing parking facilities.

2.2 Morphological Feature Characterization

This section converts spatial concentrations of bicycle parking into measurable geometric characteristics and then relates those characteristics to the surrounding road network. The procedure consists of data preprocessing and dense-core extraction, morphological quantification, and road-network-associated classification.

2.2.1 Data preprocessing and dense core extraction

Parking behaviour was represented by bicycle-order endpoints. Because operational repositioning can create a discontinuity between the endpoint of one order and the starting point of the next order for the same bicycle, repositioning events were removed before parking aggregation. For each bicycle, orders were sorted chronologically, and the displacement between the endpoint of order (i) and the starting point of the subsequent order was defined as:

$$\delta_i = \text{hav}(p_{\text{end},i}, p_{\text{start},i+1}) \quad (1)$$

where, $\text{hav}(\cdot)$ denotes the spherical haversine distance, $p_{\text{end},i}$ is the endpoint of the i -th order for a given bicycle, and $p_{\text{start},i+1}$ is the starting point of its subsequent order. If $\delta_i > 200$ m, an operational repositioning event is inferred to have occurred between the two consecutive orders, and the corresponding record is excluded from parking-event aggregation to reduce the influence of operational movements on downstream clustering.

In the dense core extraction stage, all order endpoints collected during the seven-day observation period were pre-aggregated into 10 m grids. Let the number of orders in grid (i, j) be c_{ij} , and let g denote the grid size. The order density of grid (i, j) is therefore c_{ij}/g^2 . Grids satisfying

$$\frac{c_{ij}}{g^2} \geq \theta_{\text{density}} \quad (2)$$

formed the set V . An eight-neighbourhood connected-component decomposition was applied to V to obtain candidate clusters, and components containing fewer than θ_{min} grid cells were removed to yield dense parking cores. Here, $g = 10$ m, $\theta_{\text{density}} = 0.2$ orders/ m^2 , and $\theta_{\text{min}} = 5$ grid cells. With the eight-neighbourhood definition, the maximum centre-to-centre distance between adjacent grids is $\sqrt{2}g \approx 14.14$ m, which is spatially comparable to a DBSCAN neighbourhood radius of approximately 14 m. The resulting dense cores were used as fixed spatial units for subsequent morphological and temporal analyses.

2.2.2 Morphological feature quantification

Each extracted dense core consists of a spatially connected set of parking points. Cluster membership alone does not indicate how that accumulation occupies space. A compact cluster located within a designated area and an elongated cluster extending across roadside or pedestrian space may contain similar numbers of bicycles but have very different management implications. Morphology was therefore quantified using the major and minor spatial axes of each cluster.

(1) Estimation of major-axis direction. For each cluster c , the parking-point coordinates are represented as $p_i = (x_i, y_i)$ in a local planar coordinate system, and their centroid is denoted by $\bar{p}_c$. Let n_c denote the number of parking points, equivalently the number of orders, associated with cluster c over the observation period. The covariance matrix is then constructed as:

$$\Sigma_c = \frac{1}{n_c} \sum_{i=1}^{n_c} (p_i - \bar{p}_c)(p_i - \bar{p}_c)^T \quad (3)$$

Eigendecomposition yields eigenvalue-eigenvector pairs (λ_1, u_1) and (λ_2, u_2) , with $\lambda_1 \geq \lambda_2$. The eigenvector u_1 corresponding to λ_1 represents the direction of maximum variance and is taken as the major-axis direction, while u_2

is taken as the orthogonal minor-axis direction. The major-axis azimuth is defined as:

$$\theta = \text{mod} [\text{atan} 2 (u_{1x}, u_{1y}), 180^\circ] \quad (4)$$

where, u_{1x} and u_{1y} are the eastward and northward components of the major-axis eigenvector, respectively. The azimuth θ is measured clockwise from true north and mapped to $[0^\circ, 180^\circ)$ because the PCA major axis is undirected.

(2) Measurement of major- and minor-axis spread. After determining the major- and minor-axis directions, each parking point is projected onto both axes:

$$d_{l,i} = (p_i - \bar{p}) \cdot u_1, d_{s,i} = (p_i - \bar{p}) \cdot u_2 \quad (5)$$

where, $d_{l,i}$ and $d_{s,i}$ are the projected coordinates of parking points on the major and minor axes, respectively. To reduce sensitivity to outliers, a robust spread measure based on the 5th–95th percentile range is adopted, defined as the difference between the 95th and 5th percentiles of the projected coordinates:

$$s_L = P_{95}(d_l) - P_5(d_l), s_S = \max [P_{95}(d_s) - P_5(d_s), 5 \text{ m}] \quad (6)$$

where, s_L is the major-axis spread, representing the effective longitudinal extent (m) of the cluster along its main extension direction; s_S is the minor-axis spread, representing the effective lateral bandwidth (m) of the cluster. A lower bound of 5 m was imposed on s_S to avoid unstable length-width ratios for clusters with very small minoraxis spread. s_S serves as a geometric proxy for potential lateral occupation of roadside or pedestrian space.

(3) Morphological feature summary and geometric interpretation. For each dense core, four morphological features are obtained: major-axis spread s_L , bandwidth s_S , length-width ratio ρ

$$\rho = \frac{s_L}{s_S} \quad (7)$$

and major-axis azimuth θ . Their geometric interpretation is as follows: $\rho \approx 1$ indicates an approximately isotropic cluster (i.e., a compact or nearly square morphology, such as a DPA configuration); $\rho \geq 3$ indicates a significantly elongated cluster, such as roadside band-shaped parking; θ indicates the orientation of the major axis; and s_S indicates the lateral occupation depth.

Roadside parking may exhibit a multi-row parallel arrangement. For descriptive interpretation, assuming an approximate lateral occupation depth of 1.8 m per parking row, the approximate number of parallel parking rows (k) is estimated as:

$$k = \text{round} \left(\frac{s_S}{1.8 \text{ m}} \right) \quad (8)$$

$k \approx 1$ corresponds to single-row parallel parking, whereas $k \geq 3$ indicates potential multi-row parking. Accordingly, band-shaped clusters are further subdivided into narrow-band clusters ($s_S < 10$ m, typically corresponding to approximately 3–4 rows) and wide-band clusters ($s_S \geq 10$ m, typically corresponding to approximately 6 or more rows). These categories represent relatively shallow multi-row occupation and greater lateral occupation, respectively. The estimated row count k is used as a geometric proxy for potential multi-row occupation rather than as a direct physical count of bicycle rows.

2.2.3 Road-network-associated morphology classification

The morphological interpretation of a parking band, including whether it is associated with the roadside and whether its major axis is aligned with the road, is defined relative to the surrounding road network. To this end, two road-network association indicators are introduced. The first is the shortest distance from the cluster centroid to the nearest road centreline:

$$d_{\text{road}} = \min_r \text{dist}(\bar{p}, r) \quad (9)$$

where, $\text{dist}(\bar{p}, r)$ is the shortest distance from the cluster centroid to the centreline of road segment r , and d_{road} is the minimum distance to the nearest road centreline.

The second is the orientation consistency between the cluster major axis and the alignment of the nearest road:

$$\varphi = |\cos(\theta - \theta_{\text{road}})| \in [0, 1] \quad (10)$$

where, θ_{road} denotes the alignment azimuth of the nearest road. Both θ and θ_{road} are expressed using the same angular reference system, with true north defined as 0° . The orientation-consistency index φ approaches 1 when the cluster major axis is parallel to the road alignment and approaches 0 when it is perpendicular to the road. In addition, $d_{\text{road},2}$ denotes the distance from the cluster centroid to the second-nearest road centreline.

Morphology classification does not use any POI/AOI information. Based solely on geometric features and roadnetwork association indicators, each cluster is assigned to one of four morphology types using the priority order: $RB \rightarrow IC \rightarrow DPA \rightarrow SP$. Once a cluster satisfies a higher-priority criterion, it is not evaluated against lower-priority categories.

$$RB \Leftrightarrow \rho \geq 3 \wedge d_{road} \leq 5 \text{ m} \quad (11)$$

$$IC \Leftrightarrow d_{road,2} \leq 10 \text{ m} \wedge \min(|\theta_{road,1} - \theta_{road,2}|, 180^\circ - |\theta_{road,1} - \theta_{road,2}|) \geq 30^\circ \quad (12)$$

$$DPA \Leftrightarrow \rho < 1.5 \wedge 10 \text{ m} \leq s_L < 30 \text{ m} \wedge 5 \text{ m} \leq s_S < 20 \text{ m} \quad (13)$$

where, the symbols $\vee$ and $\wedge$ represent logical “OR” and “AND”, respectively.

Clusters that do not satisfy the criteria for RB, IC, or DPA are classified as SP. RB represents elongated clusters located in close proximity to the road network. IC clusters are characterised by their proximity to two nearby roads with substantially different alignment directions. DPA clusters are identified solely from geometric characteristics and represent compact, moderately sized parking-area-like configurations, rather than being defined by POI or AOI labels. This geometry-based definition is adopted because Amap’s AOI classification does not contain a dedicated category for non-motorised vehicle parking, while generic “parking lot” POIs may also refer to facilities intended for motor vehicles.

Orientation consistency is used as a diagnostic characteristic of RB clusters rather than as a mandatory classification condition. After RB clusters are identified, $\varphi \geq \cos 30^\circ \approx 0.87$ is used to distinguish clusters whose major axes are aligned with the adjacent road from those with oblique or near-perpendicular orientations. This allows the empirical analysis to assess, rather than assume, whether roadside bicycle accumulation generally follows the road alignment.

2.3 Identification of Short-Duration Demand-Surge Clusters

Urban parking pressure is often produced not by sustained daily demand but by rapid accumulation during relatively short commuting periods. This is particularly relevant around metro entrances, employment districts, and residential areas, where large numbers of bicycles can arrive or depart within tens of minutes. Total order volume alone cannot adequately distinguish these temporally concentrated events.

Temporal information from the orders associated with each dense core, together with surrounding facility information, was then incorporated into the morphology-labelled cluster database. The analysis consisted of two steps.

(1) Peak-period accumulation intensity calculation. Let $n_{m,c}$ and $n_{e,c}$ denote the numbers of orders associated with cluster c during the morning peak (07:00–10:00) and evening peak (17:00–20:00), respectively. Using the total order volume n_c defined above, the morning- and evening-peak order shares are defined as:

$$s_{m,c} = \frac{n_{m,c}}{n_c}, s_{e,c} = \frac{n_{e,c}}{n_c} \quad (14)$$

Cluster c is classified as a short-duration demand-surge cluster if either

$$s_{m,c} \geq 0.4 \vee s_{e,c} \geq 0.4 \quad (15)$$

This criterion indicates that at least 40% of the observed activity of cluster c is concentrated within one of the defined 3-hour commuting-peak windows.

(2) Locational attribute analysis. For the identified short-duration demand-surge clusters, facility data were incorporated to characterise their locational context. The distance from the cluster centroid to the nearest metro entrance, d_{metro} , was used to identify metro-feeder locations, with locations satisfying $d_{metro} \leq 200 \text{ m}$ classified as the metro-feeder type. The AOI category containing the cluster centroid was used to characterise the surrounding functional zone, including residential, office, commercial, school, and hospital areas. Peak timing was then interpreted jointly with the surrounding functional context to distinguish commuting-related tidal roles, including morning arrival patterns around employment or transport-hub areas and evening return patterns around residential areas. These locational attributes were used to interpret the spatial context of short-duration concentrated demand and to inform subsequent management recommendations.

2.4 Governance Priority Assessment and Differentiated Management Recommendations

The morphology analysis identifies how bicycles occupy space, while the peak-demand analysis identifies when and where concentrated accumulation occurs. These two dimensions were assessed jointly to distinguish locations with high usage from locations where intensive demand coincides with parking forms that are more likely to interfere with the use of street and public space.

(1) Parking accumulation impact assessment. High bicycle demand does not necessarily imply a high management impact. Large numbers of bicycles may be accommodated with limited disturbance when parking occurs within a designated area, whereas elongated roadside accumulation may create substantial spatial pressure at a lower demand level. An estimated spatial occupation impact index I_c was therefore developed to represent the potential disturbance associated with the morphology and location of cluster c :

$$I_c = f_{\text{shape},c} \times f_{\text{width},c} \quad (16)$$

where, $f_{\text{shape},c}$ is the shape-and-road-proximity factor and $f_{\text{width},c}$ is the lateral-bandwidth factor.

For the demand-impact ($D \times I$) evaluation, the demand dimension D_c is represented by the total order volume of cluster c , such that $D_c = n_c$. An order-volume threshold of $D_0 = 500$ orders and an impact threshold of $I_0 = 0.4$ are used to divide the $D_c \times I_c$ space into four quadrants: Q1 (high demand $\times$ high impact) represents the double-high quadrant and the main focus of management; Q2 (high demand $\times$ low impact) represents locations where capacity-oriented intervention may be appropriate; Q3 (low demand $\times$ high impact) represents locations requiring spatial remediation or source guidance; and Q4 (low demand $\times$ low impact) comprises locations subject to routine observation.

(2) Construction of governance priority criterion. The short-duration demand-surge criterion and morphology criterion are combined. To support proactive remediation and dispatch, a governance priority criterion is constructed:

$$\text{Priority}(c) \Leftrightarrow (s_{m,c} \geq 0.4 \vee s_{e,c} \geq 0.4) \wedge n_c \geq 500 \wedge M_c \in \{\text{band, narrow-band}\} \quad (17)$$

where, $\text{Priority}(c)$ denotes the priority-governance status of cluster c ; n_c is the total order volume of cluster c during the 7-day observation period; and M_c denotes whether cluster c , belongs to the band or narrow-band morphology defined in Section 2.2.2.

This criterion requires the simultaneous satisfaction of three conditions: (i) activity is highly concentrated within either the morning or evening peak window ($s_{m,c} \geq 0.4$ or $s_{e,c} \geq 0.4$); (ii) the total order volume reaches the predefined high-demand threshold ($n_c \geq 500$); and (iii) the cluster exhibits a band-shaped morphology associated with greater potential lateral occupation.

(3) Differentiated governance recommendations. The resulting priority-point list was subsequently combined with AOI/POI proximity information and road-hierarchy data. Facilities considered within a 200 m radius included metro entrances, designated bicycle parking facilities, commercial facilities, bus stops, schools, and hospitals. These characteristics were used to distinguish different locational contexts among the identified priority points.

Management recommendations were then linked to these contexts. Depending on peak timing, surrounding land use, transport facilities, and road hierarchy, possible responses included pre-peak bicycle repositioning, targeted remediation of roadside accumulation, guidance toward designated parking areas, and adjustment or improved use of existing parking facilities. In this way, the analytical framework moved from hotspot identification to location-specific decision support for shared bicycle parking and urban public-space management.

3 Case Study: Empirical Analysis of Guangzhou

3.1 Data Overview and Identification Scope

Three types of data were used in the empirical analysis. First, shared bicycle order and trajectory data covered seven consecutive days from June 22 to 28, 2026. After duplicate records and observations with missing values were removed and the data were restricted to the Guangzhou administrative boundary, the dataset contained 6.902 million orders. Each record included origin and destination coordinates, start and end times, and vehicle identifiers in the GCS-02 coordinate system (also known as the Mars coordinate system). These data provided the basis for parking-morphology characterization and demand identification.

Second, the urban road-network dataset contained approximately 250,000 road segments, together with information on road hierarchy and alignment azimuth. These data were used to relate parking-cluster geometry to the surrounding street network and to support morphology classification.

Third, POI and functional-zone AOI data were used to describe the urban context of the identified clusters. The dataset included metro entrances, bus stops, residential areas, office buildings, commercial districts, schools, hospitals, and other urban facilities. These data supported the interpretation of locational characteristics and the subsequent classification of priority management locations.

Two identification scopes were used to distinguish broad geometric patterns from more restrictive road-associated morphology (Table 1). The full dataset contained 27,687 dense cores derived from 6.902 million orders and was used for the initial band/non-band classification. The strict subset contained 26,142 cores and 6.723 million orders for which the required road-network association information was available, and was used to distinguish RB, IC, DPA, and SP morphologies.

Table 1. Comparison of the two identification scopes

Scope	Time Window	Order Volume	Clustering Method	Spatial Resolution	Number of Clusters	Morphology Criterion
Full dense cores	Seven-day observation period	6.902 million	Grid + 8-neighbourhood connected components	10 m grid; maximum adjacent-cell centre distance ≈ 14.14 m	27,687	Band vs. non-band (geometry only)
Strict core subset	Seven-day observation period	6.723 million	Grid + 8-neighbourhood connected components	10 m grid; maximum adjacent-cell centre distance ≈ 14.14 m	26,142	RB/IC/DPA/SP four types (with road criteria)

Note: RB = roadside-band; IC = intersection-corner; DPA = designated-parking-area-like; SP = scattered points.

3.2 Results of Parking Hotspot Morphological Feature Characterization

3.2.1 Overall concentration of parking demand

The full-scope analysis identified 27,687 dense parking cores. Parking activity was strongly concentrated within a relatively small proportion of these locations. The top 1% of clusters by size, corresponding to 277 clusters, accounted for 43.7% of all within-cluster orders. The top 5% accounted for 66.7%, while the top 20% accounted for 85.0%. The distribution therefore indicates that a limited number of locations carried a disproportionate share of observed parking activity.

3.2.2 Morphology dichotomy in the full scope

Using geometric morphology alone (ρ and s_S), the 27,687 dense cores were divided into band-shaped (wide-band and narrow-band) and non-band-shaped clusters (Table 2 and Figure 2). Band-shaped clusters accounted for 13.43% of all cores but 32.15% of within-cluster orders. Their mean order volume was 596.5 orders per cluster, compared with 195.4 for non-band-shaped clusters, a ratio of 3.05.

Table 2. Morphology dichotomy statistics of the 27,687 dense cores

Morphology	Number of Clusters	Cluster Share	Number of Orders	Order Share	Mean Orders per Cluster
Band-shaped (wide-band and narrow-band)	3,719	13.43%	2,218,599	32.15%	596.5
Non-band-shaped clusters	23,968	86.57%	4,683,132	67.85%	195.4
Total	27,687	100.00%	6,901,731	100.00%	249.3

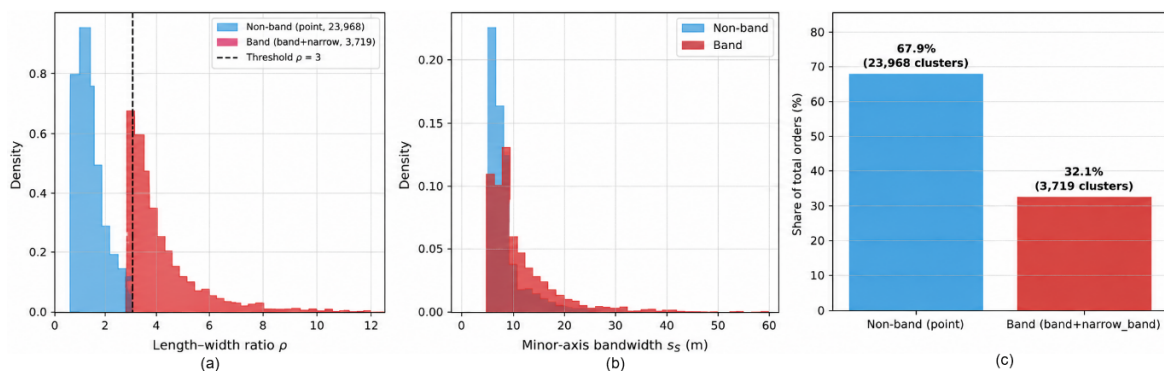


Figure 2. Morphological characteristics of the full-scope dense cores: (a) distribution of the length–width ratio ρ ; (b) distribution of the minor-axis bandwidth s_S ; (c) share of total orders for non-band and band-shaped clusters

Note: Band-shaped clusters include wide-band and narrow-band clusters.

This distribution shows that elongated parking forms were relatively uncommon in numerical terms but were associated with substantially greater activity at the cluster level. From a street-space management perspective, this combination is important because locations with elongated parking morphology account for a much larger share of bicycle activity than their numerical frequency would suggest.

3.2.3 Four-type morphology classification in the strict scope

After road-network criteria were incorporated, 26,142 cores were classified into four morphology types (Table 3 and Figure 3). The RB category contained 844 clusters, representing 3.23% of the strict sample but 10.04% of its orders. These clusters also recorded the highest mean order volume among the four morphology types. The DPA category contained 1,766 clusters and displayed comparatively compact geometry, with $\rho = 1.33$, $s_L = 14.4$ m, and $s_S = 10.9$ m. These geometric characteristics were consistent with the compact parking configurations observed in campus and park open spaces. The IC category contained 489 clusters, while the SP category contained 23,043 clusters, accounting for 88.15% of the strict sample.

Table 3. Four-type morphology statistics of the 26,142 strict cores

Morphology	Number of Clusters	Cluster Share	Number of Orders	Order Share	s_L (m)	s_S (m)	ρ	d_{road} (m)
RB	844	3.23%	675,190	10.04%	66.3	15.0	4.42	2.2
IC	489	1.87%	210,739	3.13%	28.8	14.4	1.87	3.8
DPA	1,766	6.76%	167,325	2.49%	14.4	10.9	1.33	52.3
SP	23,043	88.15%	5,669,381	84.33%	21.7	10.1	2.02	58.1
Total	26,142	100.00%	6,722,635	100.00%	—	—	—	—

Note: RB = roadside-band; IC = intersection-corner; DPA = designated-parking-area-like; SP = scattered points; s_L = longitudinal spread along the major axis; s_S = lateral spread along the minor axis; ρ = length–width ratio (s_L/s_S); d_{road} = distance from the cluster centroid to the nearest road centreline; “—” = not applicable.

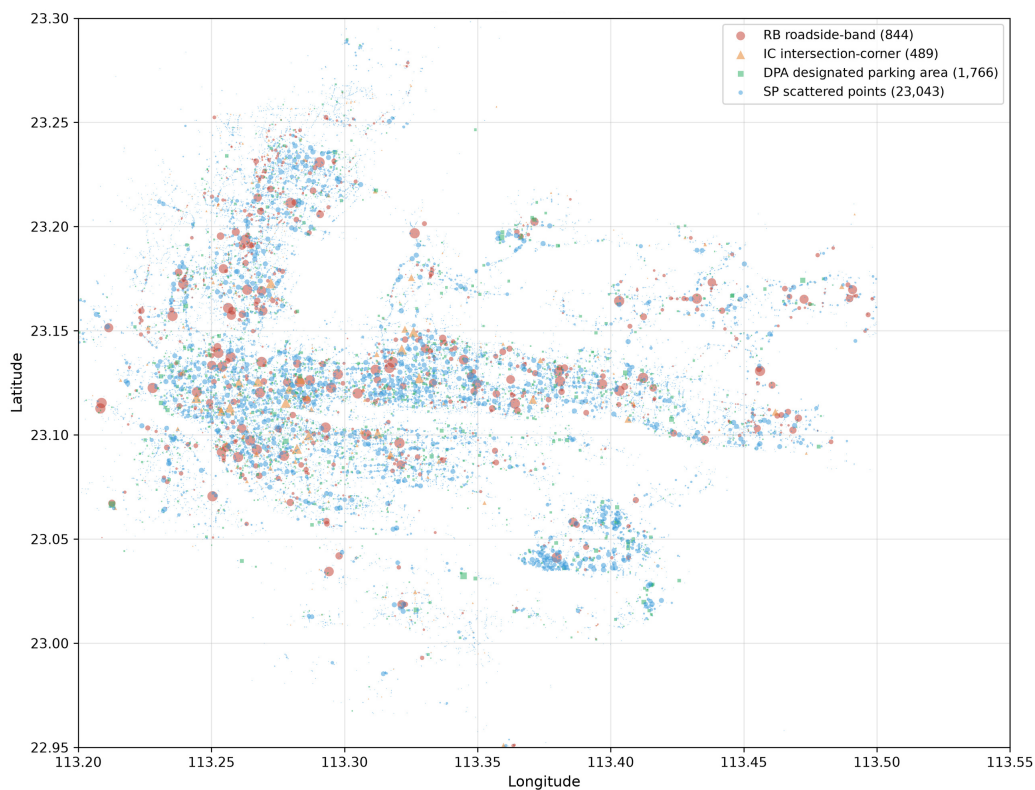


Figure 3. Spatial distribution of the four morphology types in central Guangzhou

Note: RB = roadside-band; IC = intersection-corner; DPA = designated-parking-area-like; SP = scattered points.

The four-type classification therefore separates the dominant dispersed form of bicycle parking from a smaller set of spatially distinctive clusters whose geometry and relationship with the road network have different implications for parking and public-space management.

The reduction from 13.43% band-shaped clusters in the full scope to 3.23% RB clusters in the strict scope resulted from the additional road-proximity requirement. The full-scope classification was based solely on cluster geometry and therefore included elongated clusters located away from roads. By contrast, the strict RB definition required $d_{road} \leq 5$ m, restricting this category to elongated clusters directly associated with the street network.

3.2.4 Orientation consistency and multi-row arrangement

Orientation consistency was examined for the 844 RB clusters. Only 21.8% had a major-axis-to-road angle of no more than 30° ($\varphi \geq 0.87$), while the median angle was 62.8° . Most RB clusters therefore did not extend parallel to the adjacent road. Instead, many displayed oblique or near-perpendicular orientations, consistent with bicycles being arranged in multiple rows extending into the lateral depth of roadside or pedestrian space.

This result challenges the assumption that roadside bicycle accumulation can be represented primarily as a linear pattern parallel to the road. It also explains why orientation consistency was retained as a diagnostic measure rather than imposed as a mandatory RB classification condition: requiring both proximity and parallel alignment would have excluded nearly 80% of the empirically identified roadside-band clusters.

The median minor-axis bandwidth of narrow-band clusters was 6.4 m, corresponding to an estimated 3–4 rows, whereas the median bandwidth of wide-band clusters was 12.9 m, corresponding to approximately seven rows based on the geometric proxy defined in Eq. (8). The observed roadside accumulation was therefore consistent with substantial lateral occupation rather than a single row of bicycles aligned with the curb.

3.2.5 Geometric comparison between roadside-band and designated-parking-area-like

The geometric distributions of RB and DPA clusters differed clearly (Figure 4). The length–width ratio of RB clusters was concentrated between 3 and 8, reflecting their elongated geometry, while their mean bandwidth was approximately 15 m. In contrast, DPA clusters had length–width ratios concentrated between 1.0 and 1.5 and were therefore considerably more compact. Their mean bandwidth was approximately 11 m, consistent with the compact geometry used to define the DPA category. RB clusters also recorded a substantially higher mean order volume than DPA clusters, at approximately 800 versus 95 orders per cluster, respectively. The comparison shows that RB and DPA clusters differ not only in geometric form but also in activity intensity. These differences provide an important basis for the subsequent assessment of demand and estimated spatial impact.

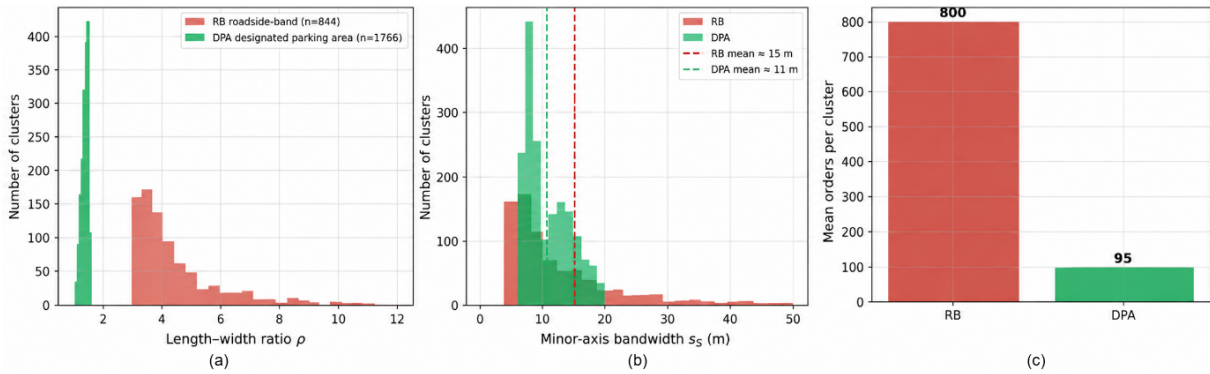


Figure 4. Comparison of RB and DPA clusters: (a) distribution of the length–width ratio ρ ; (b) distribution of the minor-axis bandwidth s_s ; (c) mean number of orders per cluster

Note: RB = roadside-band; DPA = designated-parking-area-like; ρ = length–width ratio; s_s = minor-axis bandwidth.

3.3 Short-Duration Demand Surges and Parking Impact

3.3.1 Temporal structure of high-demand clusters

Using $n_c \geq 500$ orders as the high-demand threshold, 1,842 clusters were identified. Their temporal distribution showed pronounced commuting peaks. A total of 770 clusters (41.8%) had their highest hourly activity at 08:00, while 462 clusters (25.1%) peaked at 18:00 (Figure 5). Together, the two periods accounted for approximately two-thirds of the high-demand clusters, indicating a strong association between parking concentration and commuting periods.

Applying the short-duration demand-surge criterion identified 320 morning-peak clusters ($s_{m,c} \geq 0.4$), with a median of 897 orders per cluster, and 80 evening-peak clusters ($s_{e,c} \geq 0.4$), with a median of 993 orders per cluster (Table 4). No high-demand cluster simultaneously satisfied both the morning- and evening-surge criteria. These locations represent the subset of high-demand clusters in which bicycle activity was particularly concentrated within a short peak-period window and are therefore especially relevant to time-sensitive parking management and dispatch.

3.3.2 Accumulation characteristics around metro entrances

Among the 1,842 high-demand clusters, 506 were located within 200 m of a metro entrance, accounting for 27.5% of the total. These clusters recorded a mean of 3,733 orders per cluster, approximately 1.7 times the mean of 2,207 orders per cluster observed at high-demand clusters outside the 200 m metro buffer. Their mean morning-peak order share was also higher, at 29.7%, compared with 25.6% for non-metro locations. These results show that high-demand

clusters near metro entrances were associated with greater order volumes and a somewhat stronger morning-peak concentration.

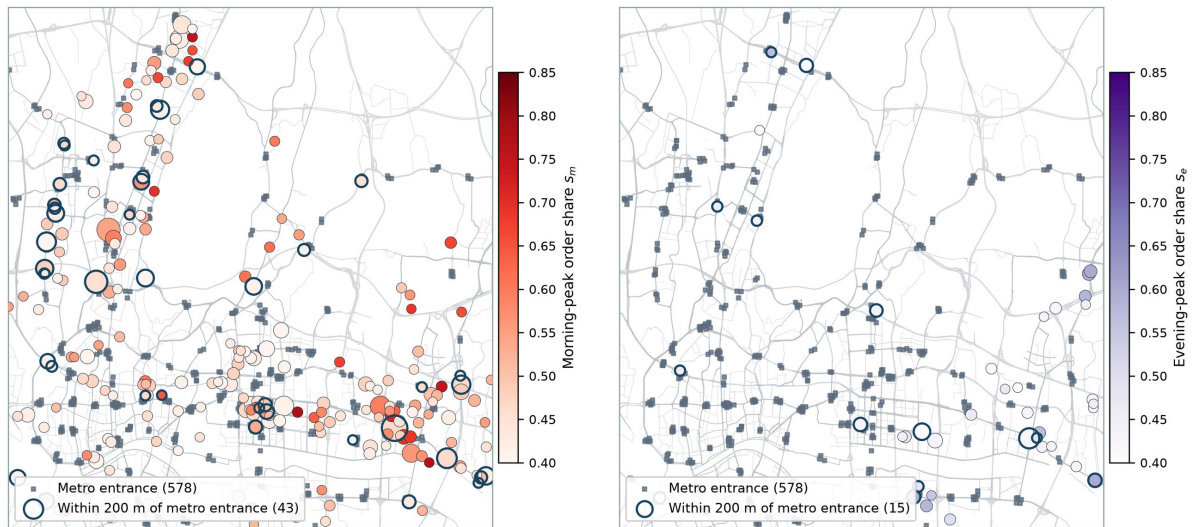


Figure 5. Spatial distribution of short-duration demand-surge clusters in central Guangzhou: (a) morning-peak surge clusters; (b) evening-peak surge clusters

Note: $s_{m,c}$ = morning-peak order share of cluster c ; $s_{e,c}$ = evening-peak order share of cluster c . Morning- and evening-peak surge clusters satisfy $s_{m,c} \geq 0.4$ and $s_{e,c} \geq 0.4$, respectively. Circles indicate surge clusters located within 200 m of a metro entrance. Counts shown in the map refer to the central-Guangzhou subset displayed in the figure.

Table 4. Period-surge and metro-proximity characteristics of the high-demand clusters ($n_c \geq 500$, 1,842 in total)

Cluster Type	Count	Median Orders per Cluster	Band Share	Count Within 200 m of Metro (Share)	Median d_{metro} (m)
Morning-peak surge ($s_{m,c} \geq 0.4$)	320	897	59.4%	72 (22.5%)	436
Evening-peak surge ($s_{e,c} \geq 0.4$)	80	993	62.5%	26 (32.5%)	566
Other high-demand clusters	1,442	—	—	—	—
All high-demand clusters within 200 m of a metro entrance	506	1,658	49.8%	506 (100.0%)	—

Note: $s_{m,c}$ = morning-peak order share of cluster c ; $s_{e,c}$ = evening-peak order share of cluster c ; “—” = not applicable.

Within the strict four-type morphology classification, SP clusters accounted for 86.6% of high-demand clusters located within 200 m of metro entrances, indicating that accumulation near stations was predominantly dispersed across the available surrounding space rather than concentrated in elongated roadside bands. This pattern is consistent with the spatial constraints around station entrances and the role of shared bicycles in first- and last-mile travel.

The distinction has direct management implications. High-demand accumulation around metro entrances was predominantly characterised by dispersed rather than elongated morphology, whereas band-shaped accumulation along arterial roads was associated with greater lateral street-space occupation. These two spatial settings therefore suggest different management priorities: capacity management and guidance towards designated areas around metro entrances, and targeted treatment of elongated roadside accumulation along major streets.

3.3.3 Demand–impact evaluation

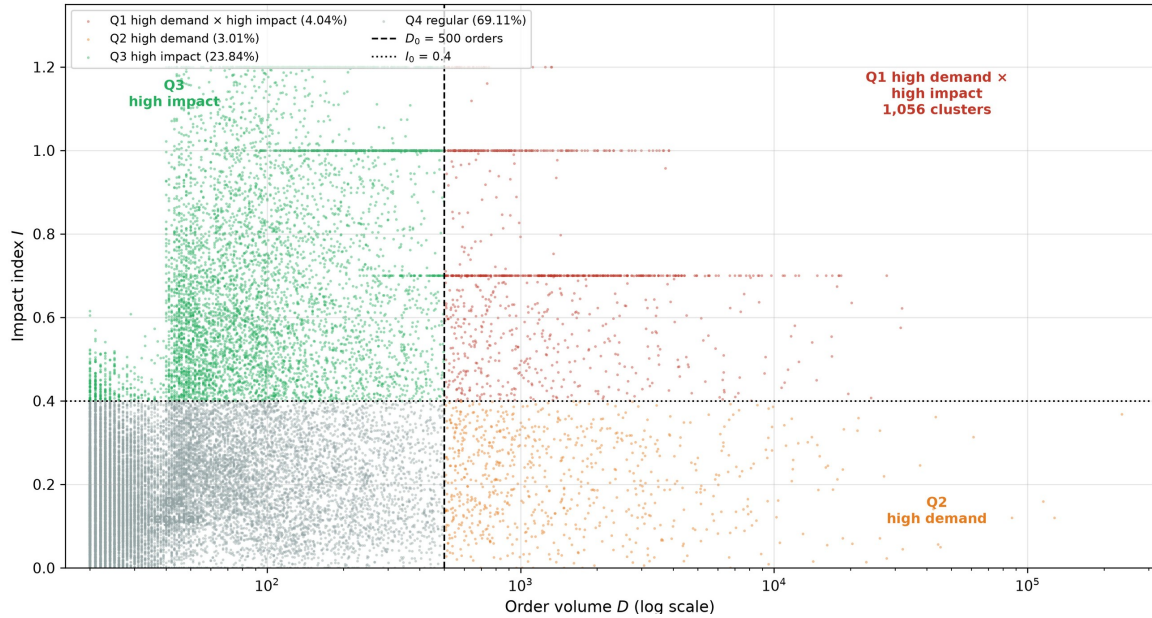
The demand–impact ($D \times I$) evaluation was applied to the 26,142 strict cores according to the procedure described in Section 2.4 (Table 5 and Figure 6). A total of 1,056 clusters, or 4.04% of the sample, fell within Q1 (high demand $\times$ high impact), accounting for 30.24% of all orders in the strict sample. Among the Q1 clusters, 910 (86.2%) were band-shaped.

Band-shaped clusters were highly represented in Q1, accounting for 86.2% of the high-demand, high-impact clusters. This pattern shows that elongated parking morphology was prominent among locations combining high demand with high estimated spatial impact. By contrast, no DPA cluster was observed in Q1, suggesting that compact DPA morphology was less likely to coincide with both high demand and high estimated spatial impact within the present classification framework.

Table 5. $D \times I$ quadrant distribution of the 26,142 strict cores

Quadrant	Clusters	Cluster Share	Orders	Order Share	Meaning
Q1 High demand $\times$ high impact	1,056	4.04%	2,033,061	30.24%	High demand and high estimated spatial impact; impact-focused intervention
Q2 High demand $\times$ low impact	786	3.01%	2,804,176	41.71%	High demand but low estimated spatial impact; capacity-oriented intervention may be appropriate
Q3 Low demand $\times$ high impact	6,233	23.84%	795,515	11.83%	Low demand but high estimated spatial impact; spatial remediation or source guidance may be appropriate
Q4 Low demand $\times$ low impact	18,067	69.11%	1,089,883	16.21%	Low demand and low estimated spatial impact; routine observation
Total	26,142	100.00%	6,722,635	100.00%	—

Note: D = demand dimension, represented by cluster total order volume ($D_c = n_c$); I = estimated spatial occupation impact index; Q1 = high demand $\times$ high impact; Q2 = high demand $\times$ low impact; Q3 = low demand $\times$ high impact; Q4 = low demand $\times$ low impact. The quadrant thresholds are $D_0 = 500$ orders and $I_0 = 0.4$. “—” = not applicable.

**Figure 6.** $D \times I$ scatter plot and quadrants of the strict cores

Note: D = demand dimension, represented by cluster total order volume ($D_c = n_c$); I = estimated spatial occupation impact index; Q1 = high demand $\times$ high impact; Q2 = high demand $\times$ low impact; Q3 = low demand $\times$ high impact; Q4 = low demand $\times$ low impact. The quadrant thresholds are $D_0 = 500$ orders and $I_0 = 0.4$.

The remaining clusters were distributed across Q2–Q4. Q2 contained 786 clusters with high demand but relatively low estimated spatial impact and accounted for 41.71% of orders. Q3 contained 6,233 clusters with low demand but high estimated spatial impact, while Q4 contained 18,067 clusters with comparatively low values on both dimensions. These differences provide a basis for distinguishing capacity-oriented management, spatial remediation, and routine monitoring.

3.4 Spatial Distribution and Locational Context of Priority Governance Points

3.4.1 Identification and spatial distribution of priority points

Application of the composite short-duration surge $\times$ high-demand $\times$ band-accumulation criterion identified 240 priority governance points across Guangzhou, including 190 morning-peak and 50 evening-peak surge points. Of these, 166 were located within the central urban area, comprising 139 morning-peak and 27 evening-peak surge points (Figure 7a).

The priority points were not evenly distributed across Guangzhou. Within the central urban area, they showed a clear tendency to follow major road corridors and to form spatial clusters. Clusters were particularly visible along the

old-city–Tianhe corridor north of the Pearl River and within the southern residential belt south of the river. Their spatial distribution broadly corresponded with major commuting corridors, linking the identified parking problem to the daily movement structure of the city.

As shown in Figure 7b, 171 of the 240 priority governance points were located in road/open-space AOIs, far exceeding the numbers in residential areas (36) and other AOI categories. Figure 7c further shows that 154 points (64%) were located within 200 m of commercial facilities and 120 (50%) within 200 m of designated parking, while smaller proportions were associated with metro entrances, hospitals, bus stops, and schools. The facility categories are not mutually exclusive, so a single priority governance point may be associated with more than one nearby facility type.

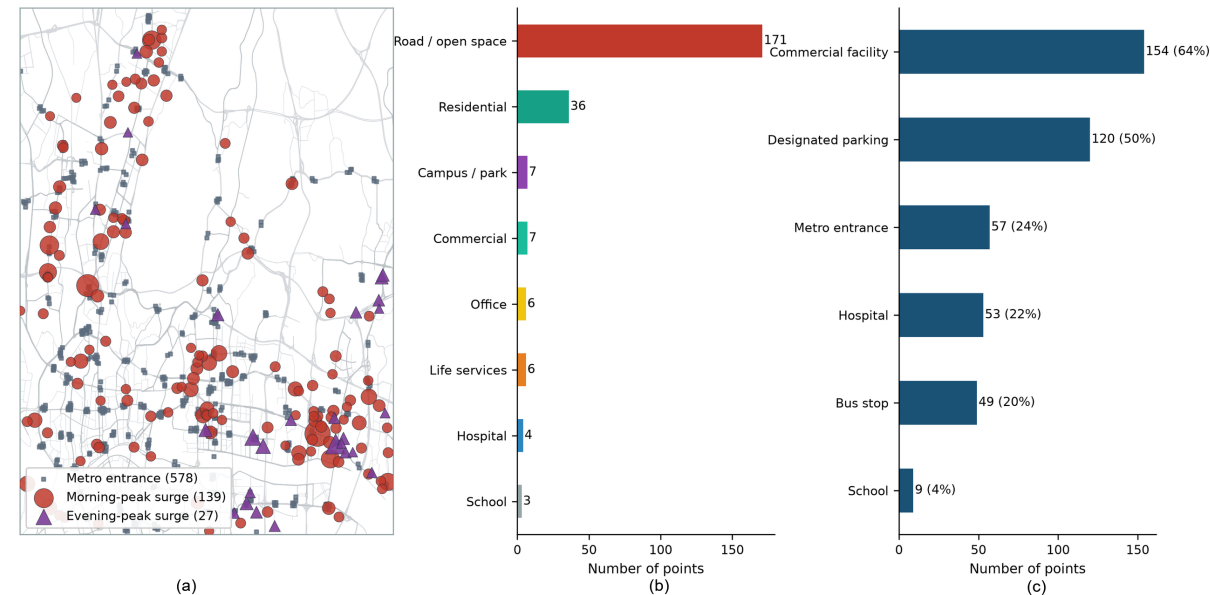


Figure 7. Spatial and locational characteristics of priority governance points: (a) spatial distribution of 166 central-area points; (b) AOI composition of all 240 points; (c) adjacent facility coverage within 200 m of all 240 points.

Note: AOI = area of interest. Facility categories in (c) are not mutually exclusive because a single priority governance point may have multiple types of facilities within a 200 m radius; therefore, the summed category counts can exceed the total number of priority governance points.

3.4.2 Relationship with area of interest, point of interest, and road hierarchy

The surrounding urban context of the 240 priority points further clarified where high-priority parking pressure occurred. In terms of AOI category, 171 points (71.3%) were located in roads and open spaces, 36 in residential areas, 7 in campus or park areas, 7 in commercial districts, 6 in office areas, 6 in living-service areas, 4 in hospital areas, and 3 in school areas (Table 6).

Facility proximity revealed an additional pattern. Within 200 m of the priority points, 64.2% were associated with commercial facilities, 50.0% with existing designated parking facilities, 23.8% with metro entrances, 22.1% with hospitals, and 20.4% with bus stops. Particularly notable is that 120 of the 240 priority points were located within 200 m of an existing designated parking facility while still displaying band-shaped accumulation.

This finding suggests that parking pressure cannot be interpreted solely as a shortage of designated facilities. At a substantial proportion of priority locations, designated parking facilities were already present nearby, yet band-shaped accumulation was still observed. For these locations, management may therefore need to address access, parking guidance, electronic-fence configuration, or operational coordination in addition to the provision of new parking capacity.

Road hierarchy showed a similarly concentrated pattern. 91 priority points were adjacent to arterial roads and 64 to sub-arterial roads, together accounting for 64.6% of the total. A further 41 were associated with branch roads, 36 with expressway service roads, and 8 with other road types. The concentration along higher-order roads reinforces the connection between priority parking locations, major urban movement corridors, and competition for limited street space.

3.4.3 Locational contexts and corresponding management responses

Combining peak timing, AOI/POI relationships, and road hierarchy revealed three recurring locational contexts among the 240 priority points. These contexts are not necessarily mutually exclusive.

Table 6. Location and facility-relationship summary of the 240 priority governance points

Dimension	Composition	Number of Points	Share
Peak type	Morning-peak surge	190	79.2%
Peak type	Evening-peak surge	50	20.8%
AOI category	Roads and open spaces	171	71.3%
AOI category	Residential areas	36	15.0%
AOI category	Commercial/office/campus or park/living services	26	10.8%
AOI category	School/hospital	7	2.9%
Facilities within 200 m	Commercial facilities	154	64.2%
Facilities within 200 m	Designated parking facilities	120	50.0%
Facilities within 200 m	Metro entrances	57	23.8%
Facilities within 200 m	Hospitals	53	22.1%
Facilities within 200 m	Bus stops	49	20.4%
Facilities within 200 m	Schools	9	3.8%
Adjacent road hierarchy	Arterial + sub-arterial	155	64.6%
Adjacent road hierarchy	Branch + expressway service road + other	85	35.4%

Note: AOI = area of interest; POI = point of interest. Facility-proximity categories are not mutually exclusive; a priority governance point may be counted in more than one facility category.

The first was the metro-feeder context, represented by 57 priority points (23.8%) located within 200 m of metro entrances. These locations were associated with transit-oriented peak-period demand and are relevant to first- and last-mile travel. Potential interventions include pre-peak capacity preparation, proactive clearance, and adjustment of feeder parking space.

The second was the arterial commuting-corridor context, represented by 155 priority points (64.6%) adjacent to arterial or sub-arterial roads. These locations were frequently associated with commercial facilities and employment-oriented areas. Their elongated roadside morphology suggests the need for roadside parking remediation, stronger guidance towards designated parking areas, and pre-peak bicycle dispatch.

A third recurring context involved priority points located within or adjacent to residential areas. At these locations, evening accumulation may be associated with return commuting, suggesting the need for clearer internal parking demarcation and the removal or redistribution of redundant bicycles during lower-demand nighttime periods. These recurring contexts highlight differences in peak timing, surrounding urban function, road environment, and corresponding management needs. Because the contexts are not necessarily mutually exclusive, individual priority points may require combinations of these management responses.

4 Conclusions and Future Work

This study developed a morphology-aware framework for identifying shared bicycle parking hotspots and translating spatial clustering results into information relevant to urban public-space management. By combining PCA, a robust spread measure based on the 5th–95th percentile range, road proximity, and orientation-consistency criteria, the framework distinguished different forms of bicycle accumulation while keeping parking morphology separate from short-duration demand intensity.

The empirical analysis of 6.902 million orders collected over one week in Guangzhou identified 27,687 dense parking cores. Parking activity was highly concentrated, with the top 1% of clusters accounting for 43.7% of within-cluster orders. Band-shaped clusters represented only 13.4% of all cores but accounted for 32.1% of orders, showing that elongated parking forms carried a disproportionately large share of activity. Within the strict four-type classification, roadside-band clusters accounted for 3.23% of the strict-core subset. Only 21.8% of these clusters were aligned within 30° of the adjacent road, while the median alignment angle was 62.8°, suggesting that oblique and potential multi-row occupation was more common than simple linear parking parallel to the curb.

Clear temporal differences were also observed. Among the 1,842 high-demand clusters, 41.8% reached their peak hourly activity at 08:00, while 320 and 80 clusters met the morning- and evening-surge criteria, respectively. High-demand clusters located within 200 m of metro entrances recorded 1.7 times the mean order volume of comparable non-metro locations, highlighting the association between metro proximity, first- and last-mile travel contexts, and local parking pressure. When demand intensity and estimated spatial impact were considered jointly, band-shaped clusters accounted for 86.2% of the high-demand, high-impact quadrant.

The composite short-duration surge $\times$ high-demand $\times$ band-accumulation criterion identified 240 priority governance points. Of these, 64.6% were located adjacent to arterial or sub-arterial roads, and 50.0% were within 200 m of existing designated parking facilities. This pattern suggests that parking pressure cannot be addressed solely by

increasing facility provision. At many priority locations, management may also need to consider parking guidance, electronic-fence configuration, access to existing facilities, and the timing of bicycle redistribution. The absence of DPA clusters from the high-demand, high-impact quadrant suggests that compact DPA morphology was less likely to coincide with both high demand and high estimated spatial impact within the present analytical framework.

These findings provide direct implications for urban mobility and public-space management. The results show that parking intensity alone is insufficient for identifying locations that require intervention; the spatial form, timing, and urban context of accumulation also matter. The distinction among morphology types allows locations to be treated differently according to how bicycle accumulation occupies street and public space. Likewise, separating sustained demand from short-duration surges provides a basis for shifting some management activities from post-hoc clearance towards pre-peak dispatch and more targeted use of existing parking facilities. Linking hotspot morphology with surrounding land use, transport facilities, and road hierarchy also supports more location-specific responses rather than uniform citywide measures.

Several limitations should nevertheless be recognised. First, the empirical analysis was based on a one-week observation period and therefore represents a short-term snapshot of parking conditions. Seasonal variation, weather effects, holidays, and longer-term changes in travel behaviour were not captured. Second, the present framework identified spatial morphology at a fixed observation scale and did not model the formation, persistence, and dissipation of individual parking hotspots over time. Third, although electric bicycles create substantial competition for non-motorised road and parking space in Guangzhou, their interaction with shared bicycle accumulation was not explicitly incorporated into the analysis. In addition, several classification and screening thresholds were specified using fixed values, and their stability across alternative parameter settings remains to be examined systematically.

Future research should therefore extend the analysis to multi-period datasets covering different seasons and travel conditions, track the temporal evolution of hotspot morphology, and examine the interaction between shared bicycles and electric bicycles within constrained urban street space. Further work should also test the robustness of the proposed classification and governance-priority rules under alternative threshold settings and across different urban contexts. Such extensions would help determine the transferability of the framework beyond Guangzhou and clarify how morphology-aware parking analysis can support more adaptive forms of urban mobility and public-space management.

Author Contributions

Conceptualization, D.X.W. and N.F.Z.; methodology, D.X.W., L.Z., and Y.D.; software, L.Z. and Y.D.; validation, M.X.Y., Y.J.S., and X.Y.L.; formal analysis, D.X.W. and L.Z.; investigation, M.X.Y., Y.J.S., X.Y.L., and L.P.; resources, L.H.Z. and N.F.Z.; data curation, D.X.W., M.X.Y., and L.P.; writing—original draft preparation, D.X.W.; writing—review and editing, D.X.W., L.Z., Y.D., L.H.Z., and N.F.Z.; visualization, D.X.W. and L.Z.; supervision, N.F.Z.; project administration, N.F.Z.; funding acquisition, N.F.Z. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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